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Hole saw

Abstract

A hole saw is movable about a rotational axis in a cutting direction to cut into a workpiece. The hole saw includes a cylindrical body with a sidewall having a first end and a second end opposite the first end, a gullet formed in the sidewall between a leading surface, a trailing surface, and a bottom surface of the sidewall, and a support aperture formed in the sidewall. The gullet extends through the first end of the body. The support aperture is entirely surrounded by the sidewall. A majority of the support aperture circumferentially overlaps with the gullet. At least a portion of the support aperture is positioned closer than the bottom surface of the sidewall to the second end. The hole saw includes a cutting tooth coupled to the first end of the sidewall adjacent the gullet and an end cap coupled to the second end of the sidewall.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1505897	12/1923	Kikta	N/A	N/A
2879036	12/1958	Wheeler	N/A	N/A
3049033	12/1961	Benjamin et al.	N/A	N/A
3080934	12/1962	Berscheid	N/A	N/A
3387637	12/1967	Ferguson et al.	N/A	N/A
3905718	12/1974	Luckenbill et al.	N/A	N/A
4527638	12/1984	Sanchez et al.	N/A	N/A
D282369	12/1985	de Villiers	N/A	N/A
4728231	12/1987	Kunimori et al.	N/A	N/A
4744704	12/1987	Galvefors	N/A	N/A
4802799	12/1988	Rachev	N/A	N/A
D303118	12/1988	Cox	N/A	N/A
4984944	12/1990	Pennington, Jr. et al.	N/A	N/A
5092716	12/1991	Omi	N/A	N/A
5338135	12/1993	Noguchi et al.	N/A	N/A
5451128	12/1994	Hattersley	N/A	N/A
5452971	12/1994	Nevills	N/A	N/A
5492187	12/1995	Neukirchen et al.	N/A	N/A
5597274	12/1996	Behner	N/A	N/A
5599145	12/1996	Reinauer et al.	N/A	N/A
5651646	12/1996	Banke et al.	N/A	N/A
D391974	12/1997	Brutscher	N/A	N/A
D392297	12/1997	Brutscher	N/A	N/A
5733074	12/1997	Stock et al.	N/A	N/A
5743162	12/1997	Sundström	N/A	N/A
5758561	12/1997	Curtsinger et al.	N/A	N/A
5794503	12/1997	Asada	N/A	N/A
5803677	12/1997	Brutscher et al.	N/A	N/A

5813802	12/1997	Ajimi et al.	N/A	N/A
5896800	12/1998	Curtsinger et al.	N/A	N/A
5904455	12/1998	Krenzer et al.	N/A	N/A
5931614	12/1998	Meyen et al.	N/A	N/A
5988953	12/1998	Berglund et al.	N/A	N/A
6007279	12/1998	Malone, Jr.	N/A	N/A
6021857	12/1999	Birk et al.	N/A	N/A
6065370	12/1999	Curtsinger et al.	N/A	N/A
D438219	12/2000	Brutscher	N/A	N/A
6205902	12/2000	Vuorinen	N/A	N/A
6299389	12/2000	Barazani	N/A	N/A
6551036	12/2002	Heule	N/A	N/A
6588992	12/2002	Rudolph	N/A	N/A
6599063	12/2002	Capstran	N/A	N/A
D478105	12/2002	Morton et al.	N/A	N/A
D478106	12/2002	Morton et al.	N/A	N/A
D478339	12/2002	Morton et al.	N/A	N/A
D478919	12/2002	Morton et al.	N/A	N/A
6601495	12/2002	Cranna	N/A	N/A
6623220	12/2002	Nuss et al.	N/A	N/A
6702047	12/2003	Huber	N/A	N/A
6705807	12/2003	Rudolph et al.	N/A	N/A
6786684	12/2003	Ecker	N/A	N/A
6820519	12/2003	Lefebvre	N/A	N/A
6959775	12/2004	Pedersen	N/A	N/A
6981911	12/2005	Martina	N/A	N/A
D516594	12/2005	Morton	N/A	N/A
7018145	12/2005	Mast et al.	N/A	N/A
7174823	12/2006	Cranna	N/A	N/A
7189036	12/2006	Watson	N/A	N/A
7241089	12/2006	Mast et al.	N/A	N/A
7258180	12/2006	Kersten	N/A	N/A
RE40297	12/2007	Berglund et al.	N/A	N/A
7401667	12/2007	Duscha et al.	N/A	N/A
D588175	12/2008	Morton	N/A	N/A
7658136	12/2009	Rompel et al.	N/A	N/A
D629820	12/2009	Van Ryswyk	N/A	N/A
7892235	12/2010	Ellis	N/A	N/A
D634343	12/2010	Burke, III	N/A	N/A
7997836	12/2010	Kim et al.	N/A	N/A
8092126	12/2011	Tanaka	N/A	N/A
8123443	12/2011	Khangar et al.	N/A	N/A
8157028	12/2011	Kersten	N/A	N/A
8162945	12/2011	Ellis	N/A	N/A
8172845	12/2011	Ellis	N/A	N/A
D664574	12/2011	Burke, III	N/A	N/A
8286736	12/2011	Weaver et al.	N/A	N/A
8444353	12/2012	Khangar et al.	N/A	N/A
D683772	12/2012	Wang	N/A	N/A
8475459	12/2012	Ellis	N/A	N/A

D687472	12/2012	Novak et al.	N/A	N/A
D692470	12/2012	Novak et al.	N/A	N/A
8550756	12/2012	Borschert et al.	N/A	N/A
8573907	12/2012	Kalomeris et al.	N/A	N/A
8579554	12/2012	Novak et al.	N/A	N/A
8579555	12/2012	Novak et al.	N/A	N/A
8608413	12/2012	Miyanaga	N/A	N/A
D701544	12/2013	Novak et al.	N/A	N/A
8721236	12/2013	Kazda et al.	N/A	N/A
D706845	12/2013	Richter	N/A	N/A
D708650	12/2013	Richter	N/A	N/A
D711441	12/2013	Novak et al.	N/A	N/A
9120162	12/2014	Keightley	N/A	N/A
9233424	12/2015	Grolimund et al.	N/A	N/A
9248518	12/2015	Elliston et al.	N/A	N/A
D765159	12/2015	Novak et al.	N/A	N/A
D789428	12/2016	Novak et al.	N/A	N/A
9884373	12/2017	Keightley	N/A	N/A
10058931	12/2017	Sussman et al.	N/A	N/A
D845362	12/2018	Novak et al.	N/A	N/A
D871877	12/2019	Terris et al.	N/A	N/A
10532412	12/2019	Ward	N/A	N/A
D875798	12/2019	Su	N/A	N/A
10730119	12/2019	Sawa et al.	N/A	N/A
D973733	12/2021	DuMez et al.	N/A	N/A
11559840	12/2022	Sawa	N/A	B23B 51/0466
12233467	12/2024	Sawa	N/A	B23B 51/0466
2002/0050196	12/2001	Fluhrer et al.	N/A	N/A
2002/0081165	12/2001	Hecht	N/A	N/A
2002/0137433	12/2001	Lee et al.	N/A	N/A
2003/0141115	12/2002	Britten et al.	N/A	N/A
2004/0050234	12/2003	Fluhrer et al.	N/A	N/A
2004/0065183	12/2003	Asada	N/A	N/A
2005/0188792	12/2004	Asada	N/A	N/A
2006/0112799	12/2005	Hambleton et al.	N/A	N/A
2007/0160435	12/2006	Chao	N/A	N/A
2007/0212179	12/2006	Khangar et al.	N/A	N/A
2008/0264231	12/2007	Coe et al.	N/A	N/A
2008/0267725	12/2007	Mills	N/A	N/A
2008/0298922	12/2007	Oettle	N/A	N/A
2009/0000454	12/2008	Baernthaler et al.	N/A	N/A
2009/0044674	12/2008	Neitzell	N/A	N/A
2009/0126712	12/2008	Kullmann et al.	N/A	N/A
2009/0274923	12/2008	Hall et al.	N/A	N/A
2009/0279972	12/2008	Novak et al.	N/A	N/A
2010/0054885	12/2009	Goetz et al.	N/A	N/A
2010/0126326	12/2009	Cloutier	N/A	N/A
2011/0197736	12/2010	Lizzi et al.	N/A	N/A

2011/0217133	12/2010	Ibarra et al.	N/A	N/A
2012/0230788	12/2011	Bozic et al.	N/A	N/A
2013/0149059	12/2012	Cranna	83/835	B23D 61/121
2013/0296866	12/2012	Ellis	N/A	N/A
2014/0013916	12/2013	Athad	N/A	N/A
2014/0017023	12/2013	Grollmund et al.	N/A	N/A
2014/0023446	12/2013	Piller et al.	N/A	N/A
2014/0023447	12/2013	Thom et al.	N/A	N/A
2014/0255117	12/2013	Tseng	N/A	N/A
2014/0271007	12/2013	Richter	N/A	N/A
2015/0000497	12/2014	Engvall et al.	N/A	N/A
2015/0239052	12/2014	Trumble et al.	N/A	N/A
2016/0059329	12/2015	Fendeleur et al.	N/A	N/A
2016/0166260	12/2015	Ellis	N/A	N/A
2016/0175945	12/2015	Chang	N/A	N/A
2016/0175947	12/2015	Chang	N/A	N/A
2017/0120347	12/2016	Lourenco et al.	N/A	N/A
2017/0120357	12/2016	Trautner et al.	N/A	N/A
2018/0085833	12/2017	Ward	N/A	N/A
2018/0193926	12/2017	Sawa et al.	N/A	N/A
2019/0217400	12/2018	Ding et al.	N/A	N/A
2024/0375193	12/2023	Sawa	N/A	B23B 51/0466

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
2865975	12/2006	CN	N/A
201394679	12/2009	CN	N/A
201597079	12/2009	CN	N/A
102689051	12/2011	CN	N/A
202963637	12/2012	CN	N/A
204171467	12/2014	CN	N/A
204321316	12/2014	CN	N/A
204565301	12/2014	CN	N/A
204818283	12/2014	CN	N/A
204997156	12/2015	CN	N/A
205519922	12/2015	CN	N/A
2422452	12/1974	DE	N/A
9417252	12/1994	DE	N/A
19623081	12/1997	DE	N/A
19810713	12/1998	DE	N/A
19910219	12/1998	DE	N/A
19913727	12/1998	DE	N/A
19939323	12/1999	DE	N/A
10042402	12/2000	DE	N/A
20101718	12/2000	DE	N/A
20113578	12/2000	DE	N/A
10253247	12/2003	DE	N/A
202004003900	12/2003	DE	N/A
102004003900	12/2004	DE	N/A

202005012328	12/2004	DE	N/A
202006010180	12/2005	DE	N/A
202006011952	12/2005	DE	N/A
0118806	12/1983	EP	N/A
0216064	12/1986	EP	N/A
0352215	12/1989	EP	N/A
0612575	12/1993	EP	N/A
0715919	12/1995	EP	N/A
0729801	12/1995	EP	N/A
1024247	12/1999	EP	N/A
1078706	12/2000	EP	N/A
1080859	12/2000	EP	N/A
1431511	12/2003	EP	N/A
1852202	12/2006	EP	N/A
1944107	12/2007	EP	N/A
1944108	12/2007	EP	N/A
2502693	12/2011	EP	N/A
2987577	12/2015	EP	N/A
2302808	12/1975	FR	N/A
2421022	12/1978	FR	N/A
191517961	12/1914	GB	N/A
550306	12/1942	GB	N/A
2201910	12/1987	GB	N/A
H0966411	12/1972	JP	N/A
2004268166	12/2003	JP	N/A
1433650	12/1987	SU	N/A
WO9706338	12/1996	WO	N/A
WO2007118718	12/2006	WO	N/A
WO2012159881	12/2011	WO	N/A
WO2012159947	12/2011	WO	N/A
WO2014180925	12/2013	WO	N/A
WO2016102261	12/2015	WO	N/A
WO2016102539	12/2015	WO	N/A

OTHER PUBLICATIONS

European Patent Office Search Report for Application No. 17192777.5 dated Mar. 13, 2018, 7 pages. cited by applicant
European Patent Office Search Report for Application No. 18150515.7 dated Jun. 20, 2018, 18 pages. cited by applicant
European Patent Office Search Report for Application No. 18150515.7 dated Sep. 18, 2018, 18 pages. cited by applicant
European Patent Office Extended Search Report for Application No. 20175783.8 dated Aug. 24, 2020 (16 pages). cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/065,305, filed Dec. 13, 2022, now U.S. Pat. No. 12,233,467, which is a divisional of U.S. patent application Ser. No. 16/910,166, filed Jun. 24, 2020, now U.S. Pat. No. 11,559,840, which is a divisional of U.S. patent application Ser. No. 15/863,253, filed Jan. 5, 2018, now U.S. Pat. No. 10,730,119, which claims priority to U.S. Provisional Patent Application No. 62/443,317, filed Jan. 6, 2017, the entire contents of which are incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates to hole saws, and more particularly to gullets formed in the hole saws.

SUMMARY

(2) In one aspect, a hole saw is movable about a rotational axis in a cutting direction to cut into a workpiece. The hole saw includes a cylindrical body having a sidewall with a first end and a second end opposite the first end. The cylindrical body also includes a gullet formed in the sidewall between a leading surface, a trailing surface, and a bottom surface of the sidewall. The gullet defines a central longitudinal axis extending between the first and second ends of the sidewall that is obliquely oriented relative to the rotational axis such that the gullet is angled from the first end to the bottom surface in a direction toward the cutting direction. The gullet has a dimension measured parallel to the rotational axis between the first end of the sidewall and the bottom surface of the gullet that is greater than 1.5 inches. The hole saw also includes a cutting tooth coupled to the first end of the sidewall adjacent the gullet.

(3) In another aspect, a hole saw is movable about a rotational axis in a cutting direction to cut into a workpiece. The hole saw includes a cylindrical body having a sidewall with a first end and a second end opposite the first end. The cylindrical body also includes a gullet formed in the sidewall between a leading surface, a trailing surface, and a bottom surface of the sidewall. The gullet extends through the first end of the sidewall and defines a central longitudinal axis extending between the first and second ends of the sidewall. The central longitudinal axis is obliquely oriented relative to the rotational axis such that the gullet is angled from the first end to the bottom surface in a direction toward the cutting direction. The hole saw also includes a cutting tooth coupled to the first end of the sidewall adjacent the gullet. A dimension measured between the leading surface and a tip of the cutting tooth in a direction perpendicular to the rotational axis is between about 0.5 inches and about 0.7 inches.

(4) In yet another aspect, a hole saw is movable about a rotational axis in a cutting direction to cut into a workpiece. The hole saw includes a cylindrical body having a sidewall with a first end and a second end opposite the first end. The cylindrical body also includes a gullet formed in the sidewall between a leading surface, a trailing surface, and a bottom surface of the sidewall. The trailing surface defines a portion having a first radius. The bottom surface has a second radius. The first radius is equal to or larger than the second radius. The gullet extends through the first end of the sidewall and defines a central longitudinal axis extending between the first and second ends of the sidewall. The central longitudinal axis is obliquely oriented relative to the rotational axis such that the gullet is angled from the first end to the bottom surface in a direction toward the cutting direction. The gullet has a first dimension measured parallel to the rotational axis between the first end of the sidewall and the bottom surface of the gullet that is greater than 1.5 inches. The hole saw also includes a cutting tooth coupled to the first end of the sidewall adjacent the gullet. A second dimension measured perpendicular to the rotational axis between the leading surface and a tip of the cutting tooth is between about 0.5 inches and about 0.7 inches.

(5) Other aspects of the invention will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a perspective view of a hole saw according to one embodiment of the invention coupled to a power tool.

(2) FIG. **2** is an end view of the hole saw of FIG. **1**.

(3) FIG. **3** is a side view of the hole saw of FIG. **1** in a flattened state and illustrating three gullets.

(4) FIG. **4** is a detailed view of a portion of the hole saw of FIG. **3**.

(5) FIG. **5** is a cross sectional view of the hole saw taken along line **5-5** of FIG. **2** illustrating the hole saw cutting through a workpiece.

(6) FIG. **6** is a planar side view of a hole saw according to another embodiment of the invention.

(7) FIG. **7** is a planar side view of a hole saw according to another embodiment of the invention.

(8) FIG. **8** is a planar side view of a hole saw according to another embodiment of the invention.

(9) FIG. **9** is a perspective view of a hole saw according to another embodiment of the invention.

(10) FIG. **10** is a side view of the hole saw of FIG. **9** in a flattened state and illustrating three gullets.

(11) FIG. **11** is a cross sectional view of the hole saw taken along line **11-11** of FIG. **9** with the hole saw cutting through two workpieces.

(12) FIG. **12** is a planar side view of a hole saw according to another embodiment of the invention.

(13) FIG. **13** is a planar side view of a hole saw according to another embodiment of the invention.

(14) FIG. **14** is a planar side view of a hole saw according to another embodiment of the invention.

(15) FIG. **15** is a planar side view of a hole saw according to another embodiment of the invention.

(16) FIG. **16** is a planar side view of a hole saw according to another embodiment of the invention.

(17) FIG. **17** is a planar side view of a hole saw according to another embodiment of the invention.

(18) FIG. **18** is a planar side view of a hole saw according to another embodiment of the invention.

(19) FIG. **19** is a planar side view of a hole saw according to another embodiment of the invention.

DETAILED DESCRIPTION

(20) Before any embodiments of the invention are explained in detail, it is to be understood that the invention is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The invention is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings. Further, “connected” and “coupled” are not restricted to physical or mechanical connections or couplings. Terms of degree, such as “about,” “substantially,” or “approximately” are understood by those of ordinary skill to refer to reasonable ranges outside of the given value, for example, general tolerances associated with manufacturing, assembly, and use of the described embodiments.

(21) FIGS. **1-5** illustrate a hole saw **100** including a cylindrical body **105** defining a cavity **110**. The cylindrical body **105** includes a sidewall **115** having a first end **120** and a second end **125** with the second end **125** positioned opposite the first end **120**. The hole saw **100** also includes an end cap **130** coupled to the second end **125** and has an aperture **135** sized to receive an arbor **140** such that the hole saw **100** can be coupled to and driven by a tool **145** (e.g., a power drill or the like) via the

arbor **140**. The tool **145** drives the hole saw **100** about a rotational axis **150** in a cutting direction **155** (FIG. **2**). In the illustrated embodiment, the arbor **140** is selectively coupled to the end cap **130**, but in other embodiments, the arbor **140** can be fixedly coupled to the end cap **130**. In some embodiments, a pilot drill bit **156** (e.g., a twist drill bit, a spade drill bit, etc.) can be selectively or fixedly coupled to the arbor **140** to extend through the cavity **110** and beyond the first end **120** of the sidewall **115**. The illustrated cylindrical body **105** is made of a metallic material (e.g., low carbon steel, etc.). In other embodiments, the cylindrical body **105** can be made of a different metallic material (e.g., high carbon steel, etc.).

(22) As best shown in FIG. **2**, the sidewall **115** defines a maximum outer diameter **160**. The outer diameter **160** can be between about 1 inch and about 7 inches. In some embodiments, the outer diameter **160** can be about 1.37 inches, about 2.56 inches, about 4 inches, or about 6.25 inches. In further embodiments, the outer diameter **160** can be greater than 7 inches. The sidewall **115** also defines a maximum thickness **165**. The thickness **165** can be between about 0.07 inches and about 0.11 inches. In some embodiments, the thickness **165** can be about 0.08 inches, about 0.09 inches, about 0.1 inches, or about 0.104 inches. As best shown in FIG. **3**, the sidewall **115** further defines a sidewall height or dimension **170** extending between the first end **120** and the second end **125** in a direction parallel to the rotational axis **150**. The illustrated sidewall height **170** is between about 2.5 inches and about 3 inches. In other embodiments, the sidewall height **170** can be between about 1.5 inches and about 5 inches.

(23) In the illustrated embodiment, the sidewall **115** includes three gullets **175** equally spaced about the rotational axis **150** (e.g., each gullet **175** is spaced about 120 degrees relative to an adjacent gullet **175**). In other embodiments, the hole saw **100** can include more than three gullets **175** (e.g., four, five, six, etc.) or can include less than three gullets (e.g., two or one) with the gullets **175** equally or non-equally spaced about the rotational axis **150**. For example, the hole saw **100** can include two gullets **175** (spaced 180 degrees apart) when the outer diameter **160** is less than about 2.25 inches.

(24) With reference to FIG. **3**, each gullet **175** defines a central longitudinal axis **180** extending between the first end **120** and the second end **125**. Each gullet **175** includes an open end **185** adjacent the first end **120** and is defined by a bottom surface **190** of the sidewall **115** opposite the open end **185**, a leading surface **195** of the sidewall **115** extending between the open end **185** and the bottom surface **190**, and a trailing surface **200** of the sidewall **115** extending between the open end **185** and the bottom surface **190**. The leading surface **195** and the trailing surface **200** of each gullet **175** are in relation to the cutting direction **155** of the hole saw **100** (i.e., the leading surface **195** is forward the trailing surface **200** as the hole saw **100** moves in the cutting direction **155**). Each trailing surface **200** includes a seat or notch **205** formed adjacent the open end **185** and a trailing surface portion **210** positioned between the notch **205** and the bottom surface **190** in a direction parallel to the rotational axis **150**. In particular, a first portion **215** of each trailing surface **200** extends between the notch **205** and the trailing surface portion **210** and a second portion **220** of each trailing surface **200** extends between the trailing surface portion **210** and the bottom surface **190**. In addition, an edge **225** is positioned between the leading surface **195** and the first end **120** that defines a substantially 90 degree edge. In other embodiments, the edge **225** can be a curved edge.

(25) In the illustrated embodiment, all three gullets **175** include the substantially same geometry, but in other embodiments, two or more gullets **175** can include different geometries. One of the gullets **175** will be described in detail below but can be applicable to one or more of the remaining gullets **175**. The illustrated first portion **215** of the trailing surface **200** is a substantially linear surface oriented substantially parallel to the rotational axis **150**. In addition, the central longitudinal axis **180** of the gullet **175** is parallel to the rotational axis **150** so that the first portion **215** is also parallel to the central longitudinal axis **180**. In other embodiments, the first portion **215** can be a curved surface. The illustrated trailing surface portion **210** defines a first radius **230** between about

1 inch and about 1.25 inches. In other embodiments, the first radius **230** can be between about 1.25 inches and about 2 inches or the first radius **230** can be between about 0.4 inches and about 1 inch. In addition, the trailing surface portion **210** is formed in a lower half of the gullet **175** in a direction parallel to the rotational axis **150**. The illustrated second portion **220** of the trailing surface **200** is a substantially linear surface, but in other embodiments, the second portion **220** can be a curved surface. The second portion **220** is also oriented at an oblique angle **235** relative to the rotational axis **150**. The illustrated oblique angle **235** is between about 40 degrees and about 50 degrees. In other embodiments, the oblique angle **235** can be between about 50 degrees and about 70 degrees or the oblique angle **235** can be between about 20 degrees and about 40 degrees. A first angle **240** is defined between the first portion **215** and the second portion **220**. The illustrated first angle **240** is between about 130 degrees and about 140 degrees. In other embodiments, the first angle **240** can be between 140 degrees and about 170 degrees, or the first angle **240** can be between about 80 degrees and about 130 degrees. In further embodiments, the first radius **230** of the trailing surface portion **210** can be omitted such that only the first angle **240** is formed between the first portion **215** and the second portion **220**. The illustrated bottom surface **190** defines a second radius **245** between about 0.25 inches and about 0.4 inches. In other embodiments, the second radius **245** can be between about 0.4 inches and about 1 inch, or the second radius **245** can be between about 0.1 inches and about 0.25 inches. In some embodiments, the second radius **245** can be equal to or less than the first radius **230**. A ratio of the first radius **230** to the second radius **245** is between about 2.5 to about 5. In other embodiments, the ratio of the first radius **230** to the second radius **245** is between about 1 and about 6. In further embodiments, the ratio of the first radius **230** to the second radius **245** is between about 3 and about 4.
(26) The illustrated leading surface **195** is a substantially linear surface oriented substantially parallel to the rotational axis **150**. A second angle **250** is defined between the second portion **220** and the leading surface **195**. The illustrated second angle **250** is between about 40 degrees and about 50 degrees. In other embodiments, the second angle **250** can be between 50 degrees and about 80 degrees or the second angle **250** can be between about 10 degrees and about 40 degrees. In further embodiments, the second radius **245** can be omitted such that only the second angle **250** is formed between the second portion **220** and the leading surface **195**.
(27) With continued reference to FIG. **3**, the gullet **175** defines a gullet height or dimension **255** extending between the first end **120** and the bottom surface **190** in a direction parallel to the rotational axis **150**. The illustrated gullet height **255** is greater than about 1.5 inches. In other embodiments, the gullet height **255** can be between about 1.7 inches and about 1.9 inches, or the gullet height **255** can be between about 1.9 inches and about 3 inches. As such, a ratio of the sidewall height **170** over the gullet height **255** is about 1.4. In other embodiments, the ratio of the sidewall height **170** over the gullet height **255** can be between about 1.1 and about 2. In further embodiments, the ratio of the sidewall height **170** over the gullet height **255** can be between about 2 and about 4. The illustrated sidewall **115** also defines a height **260** extending between the bottom surface **190** of each gullet **175** and the second end **125** of the sidewall **115**. The height **260** is between about 0.7 inches and about 0.9 inches. In other embodiments, the height **260** can be between about 0.9 inches and about 1.5 inches or the height **260** can be between about 0.1 inches and about 0.7 inches. The gullet **175** also defines a maximum gullet width **265** extending between the leading surface **195** and the trailing surface **200** in a direction perpendicular to the rotational axis **150**. In particular, the maximum gullet width **265** extends between the leading surface **195** and a point **270** located on the first portion **215** directly below the notch **205**. The illustrated maximum gullet width **265** is between about 0.75 inches and about 1.2 inches. In other embodiments, the maximum gullet width **265** can be between about 1.2 inches and about 1.5 inches, or the maximum gullet width **265** can be between about 0.4 inches and about 0.75 inches.
(28) The hole saw **100** also includes cutting teeth **275** positioned adjacent the first end **120** of the sidewall **115**. Each cutting tooth **275** has a cutting tip **280** positioned beyond the first end **120** (i.e.,

above the first end **120** as illustrated in FIG. **4**). In particular, the cutting teeth **275** are carbide cutting teeth that are seated within the notches **205** of the sidewall **115** and secured to the sidewall **115** (e.g., by a brazing operation or the like). The carbide cutting teeth **275** include a material hardness that is greater than a material hardness of the sidewall **115**. In other embodiments, the cutting teeth **275** can be made of a different material (e.g., hardened steel, etc.). In further embodiments, the sidewall **115** can form the cutting teeth **275** so that the cutting teeth **275** are not separately secured to the sidewall **115**. Each cutting tooth **275** includes a rake surface **285** and a relief surface **290** with each rake surface **285** facing into one of the gullets **175**. Each rake surface **285** defines a positive rake angle **295** (e.g., angled toward the cutting direction **155**) relative to the rotational axis **150** between about 5 degrees and about 15 degrees. As such, a minimum gullet width **296** between each cutting tip **280** and the corresponding leading surface **195** perpendicular to the rotational axis **150** is between about 0.5 inches and about 0.7 inches. In other embodiments, the minimum gullet width **296** can be between about 0.7 inches and about 1.2 inches. In further embodiments, the minimum gullet width **296** can be greater than the maximum gullet width **265**. In yet further embodiments, the rake angle **295** can be a negative rake angle (e.g., angled away from the cutting direction **155**). A tooth angle **300** is defined between the rake surface **285** and the relief surface **290** between about 60 degrees and about 70 degrees (FIG. **4**). As best shown in FIG. **2**, each cutting tip **280** defines a tip width **305** between about 0.1 inches and about 0.2 inches. The tip width **305** of each cutting tooth **275** is greater than the thickness **165** of the sidewall **115**. As such, a total cutting diameter of the hole saw **100** is greater than the maximum outer diameter **160** of the sidewall **115**. In other embodiments, the tip width **305** can be substantially the same as the thickness **165** of the sidewall **115** with the cutting teeth **275** bent radially inward or outward relative to the rotational axis **150**.

(29) With continued reference to FIG. **4**, a recess or relief **310** is formed within the first end **120** of the sidewall **115** directly behind each cutting tooth **275** with a trailing portion **315** of each recess **310** angled in a direction away from the cutting direction **155**. Each recess **310** defines a recess angle **320** relative to the rake surface **285** that is less than the tooth angle **300**. As such, during a grinding process to shape the cutting teeth **275** (e.g., to shape the tooth angle **300**), the recesses **310** provide enough clearance for a grinding wheel or the like to shape the cutting teeth **275** without contacting the sidewall **115**.

(30) Once the cutting teeth **275** are coupled to the sidewall **115**, each gullet **175** defines an area **325** between the rake surface **285**, the trailing surface **200**, the bottom surface **190**, and the leading surface **195**. In the illustrated embodiment, the area **325** of each gullet **175** is equal to or greater than 1.2 inches squared. In other embodiments, the area **325** of each gullet **175** is between about 1.2 inches squared and about 3.5 inches squared. In further embodiments, the area **325** of each gullet **175** is between about 1.2 inches squared and about 2 inches squared. As the hole saw **100** includes three gullets **175**, the total area of the illustrated gullets **175** is equal to or greater than 3.6 inches squared. Each gullet **175** also defines a volume, which is determined by multiplying the area **325** of each gullet **175** by the maximum thickness **165** of the sidewall **115**. As such, the volume of each gullet **175** is equal to or greater than 0.084 inches cubed. In some embodiments, a ratio of the tip width **305** over the volume of one of the gullets **175** is about 0.2. In other embodiments, the ratio of the tip width **305** over the volume of one of the gullets **175** can be between about 0.1 and about 1.1.

(31) With reference to FIG. **5**, the hole saw **100** is operable to cut into a workpiece **330** that defines a workpiece height **335**. In the illustrated embodiment, the workpiece height **335** is about 1.5 inches (e.g., height of a standard 2 by 4 piece of lumber). In other embodiments, the workpiece height **335** can be less than 1.5 inches. In operation, the cutting teeth **275** cut into the workpiece **330**, thereby forming workpiece chips that collect within the gullets **175**. As the hole saw **100** bores deeper into the workpiece **330**, more workpiece chips are collected within the gullets **175**. Accordingly, the illustrated gullets **175** include a particular volume to accommodate the workpiece

chips as the hole saw **100** cuts into the workpiece **330**. If the volume of the gullets **175** is not sufficient to accommodate the workpiece chips, the workpiece chips will build up within the gullets **175** and decrease the efficiency of the hole saw **100** cutting into the workpiece **330**. Once the hole saw **100** bores through the workpiece **330**, a cylindrical workpiece plug will be removed from the cavity **110** of the hole saw **100**. In particular, a user can grip the cylindrical workpiece plug (via their fingers) through the gullets **175** to axially slide the cylindrical workpiece plug along the rotational axis and out of the cavity **110**. The minimum gullet width **296** of each gullet **175** is sized to provide enough clearance between each cutting tip **280** and the corresponding leading surface **195** so that a user's finger avoids contact with the cutting tip **280** as the cylindrical workpiece plug is removed from the hole saw **100**.

(32) FIG. **6** illustrates a planar view of a portion of a hole saw **400** according to another embodiment. The hole saw **400** is similar to the hole saw **100** of FIGS. **1-5**; therefore, similar components are designated with similar references numbers plus 300. At least some differences and/or at least some similarities between the hole saws **100**, **400** will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers.

(33) The illustrated hole saw **400** is moveable about a rotational axis **450** in a cutting direction **455** and includes a sidewall **415** having a first end **420** and a second end **425**. The sidewall **415** also includes gullets **475** (only one gullet **475** is illustrated in FIG. **6**) each defining a central longitudinal axis **480**. Each gullet **475** includes a notch **505**, an open end **485**, a bottom surface **490** defining a second radius **545**, a leading surface **495**, and a trailing surface **500**. Each trailing surface **500** includes a trailing surface portion **510** defining a first radius **530**, a first portion **515**, and a second portion **520** with the trailing surface portion **510** positioned between the first portion **515** and the second portion **520**. In the illustrated embodiment, each first portion **515** is parallel to the leading surface **495**; however, in other embodiments, the first portion **515** can be obliquely angled relative to the leading surface **495**. In other embodiments, the second portion **520** can be omitted so that the trailing surface portion **510** directly connects with the bottom surface **490**. In the illustrated embodiment, the first portion **515** of each trailing surface **500** is oriented at a third angle **640** between about 5 degrees and about 15 degrees relative to the rotational axis **450** so that the first portion **515** is angled from the first end **420** to the second end **425** in a direction toward the cutting direction **455**. As a result, each central longitudinal axis **480** is also oriented between about 5 degrees and about 15 degrees relative to the rotational axis **450** so that the central longitudinal axis **480** is angled from the first end **420** to the second end **425** in a direction toward the cutting direction **455**. In other embodiments, the third angle **640** is greater than about 5 degrees. In further embodiments, the third angle **640** is less than about 15 degrees. In yet further embodiments, the third angle **640** is between about 1 degree and about 40 degrees.

(34) The hole saw **400** also includes cutting teeth **575** (only one cutting tooth **575** is illustrated in FIG. **6**) each having a cutting tip **580** and seated within one notch **505**. Each cutting tooth **575** includes a rake surface **585** and a relief surface **590**.

(35) FIG. **7** illustrates a planar view of a portion of a hole saw **700** according to another embodiment. The hole saw **700** is similar to the hole saw **100** of FIGS. **1-5**; therefore, similar components are designated with similar references numbers plus 600. At least some differences and/or at least some similarities between the hole saws **100**, **700** will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers.

(36) The illustrated hole saw **700** is moveable about a rotational axis **750** in a cutting direction **755** and includes a sidewall **715** having a first end **720** and a second end **725**. The sidewall **715** also includes gullets **775** (only one gullet **775** is illustrated in FIG. **7**) each defining a central

longitudinal axis **780**. Each gullet **775** includes a notch **805**, an open end **785**, a bottom surface **790** defining a second radius **845**, a leading surface **795**, and a trailing surface **800**. Each trailing surface **800** includes a trailing surface portion **810** defining a first radius **830**, a first portion **815**, and a second portion **820** with the trailing surface portion **810** positioned between the first portion **815** and the second portion **820**. Each leading surface **795** includes a third portion **945**, a fourth portion **950**, and a support surface **955** positioned between the third portion **945** and the fourth portion **950** in a direction parallel to the rotational axis **750**. The support surface **955** is also positioned at a support surface distance **960** between about 0.7 inches and about 1 inch from the first end **720** of the sidewall **715**. In other embodiments, the support surface **955** can be formed within at least one of the gullets **775**. In the illustrated embodiment, the third portion **945** is oriented parallel to the fourth portion **950** of each leading surface **795** and the first portion **815** of each trailing surface **800**. Each third portion **945** and fourth portion **950** are also offset (e.g., nonlinear) relative to each other. In other embodiments, the third portion **945** can be obliquely angled relative to the first portion **815** and/or the fourth portion **950**. Each support surface **955** is oriented at a support angle **965** relative to the rotational axis **750**. The support angle **965** is less than 90 degrees relative to the rotational axis **750** so that the support surface **955** is angled downwardly and toward the cutting direction **755** (e.g., the support surface **955** generally faces upwardly toward the first end **720**). In other embodiments, the support angle **965** can be less than 90 degrees and greater than 75 degrees relative to the rotational axis **750**. Also, each support surface **955** is positioned above the trailing surface portion **810** in a direction parallel to the rotational axis **750**. In other embodiments, each support surface **955** can be positioned at the same height of the trailing surface portion **810** or positioned below the trailing surface portion **810**.

(37) The hole saw **700** also includes cutting teeth **875** (only one cutting tooth **875** is illustrated in FIG. **7**) each having a cutting tip **880** and seated within one notch **805**. Each cutting tooth **875** includes a rake surface **885** and a relief surface **890**.

(38) In operation, a cylindrical workpiece plug can be removed from the hole saw **700** by using a tool (e.g., a screwdriver or the like). In particular, the tool is inserted within one of the gullets **775** below the cylindrical workpiece plug. In some embodiments, the tool engages the bottom surface **790** of the gullet **775** to leverage and push the cylindrical workpiece plug out of the first end **720**. The tool can also engage the support surface **955** to leverage and push the cylindrical workpiece plug out of the first end **720**. As the support surface **955** is angled downwardly and toward the cutting direction **755**, the tool can be captured on the support surface **955** without sliding into the gullet **775** as the tool is used to leverage the cylindrical workpiece plug out of the hole saw **700**.

(39) FIG. **8** illustrates a planar view of a portion of a hole saw **1000** according to another embodiment. The hole saw **1000** is similar to the hole saw **100** of FIGS. **1-5**; therefore, similar components are designated with similar references numbers plus 900. At least some differences and/or at least some similarities between the hole saws **100**, **1000** will be discussed in detail below. In addition, components or features described with respect to only one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers.

(40) The illustrated hole saw **1000** is moveable about a rotational axis **1050** in a cutting direction **1055** and includes a sidewall **1015** having a first end **1020** and a second end **1025**. The sidewall **1015** also includes gullets **1075** (only one gullet **1075** is illustrated in FIG. **8**) each defining a central longitudinal axis **1080**. Each gullet **1075** includes a notch **1105**, an open end **1085**, a bottom surface **1090** defining a second radius **1145**, a leading surface **1095**, and a trailing surface **1100**. Each trailing surface **1100** includes a trailing surface portion **1110** defining a first radius **1130**, a first portion **1115**, and a second portion **1120** with the trailing surface portion **1110** positioned between the first portion **1115** and the second portion **1120**. Each leading surface **1095** includes a third portion **1245**, a fourth portion **1250**, and a support surface **1255** (similar to the support surface **955** of FIG. **7**) formed on a protrusion **1270** of the sidewall **1015**, which extends into each gullet

1075. The illustrated third portion **1245** and fourth portion **1250** are substantially collinear surfaces with the support surface **1255** oriented at a substantially 90 degree angle relative to the rotational axis **1050**.

(41) The hole saw **1000** also includes cutting teeth **1175** (only one cutting tooth **1175** is illustrated in FIG. **8**) each having a cutting tip **1180** and seated within one notch **1105**. Each cutting tooth **1175** includes a rake surface **1185** and a relief surface **1190**.

(42) FIGS. **9-11** illustrate a hole saw **1300** according to another embodiment. The hole saw **1300** is similar to the hole saw **100** of FIGS. **1-5**; therefore, similar components are designated with similar references numbers plus 1200. At least some differences and/or at least some similarities between the hole saws **100**, **1300** will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers.

(43) The illustrated hole saw **1300** is moveable about a rotational axis **1350** in a cutting direction **1355** and includes a cylindrical body **1305** having a sidewall **1315** that defines a cavity **1310** and an end cap **1330** having an aperture **1335**. The sidewall **1315** includes a first end **1320** and a second end **1325**. The sidewall **1315** defines a maximum outer diameter **1360** (FIG. **11**), a maximum thickness **1365** (FIG. **11**), and a sidewall height or dimension **1370** (FIG. **10**) extending between the first end **1320** and the second end **1325** in a direction parallel to the rotational axis **1350**. The sidewall height **1370** can be between about 3.3 inches and about 3.7 inches. In other embodiments, the sidewall height **1370** can be between about 3.7 inches and about 4.5 inches. The sidewall **1315** also includes gullets **1375** each defining a central longitudinal axis **1380** oriented at an oblique angle (e.g., between about 5 degrees and about 15 degrees) relative to the rotational axis **1350**. Each gullet **1375** includes a notch **1405**, an open end **1385**, a bottom surface **1390** defining a second radius **1445**, a leading surface **1395**, and a trailing surface **1400**. Each trailing surface **1400** includes a trailing surface portion **1410** defining a first radius **1430**, a first portion **1415**, and a second portion **1420** with the trailing surface portion **1410** positioned between the first portion **1415** and the second portion **1420**. Each leading surface **1395** includes an edge **1425** and two support surfaces **1555***a*, **1555***b* (similar to the support surfaces **955**, **1255** illustrated in FIGS. **7** and **8**) so that each leading surface **1395** includes a third portion **1545**, a fourth portion **1550**, and a fifth portion **1575**. With the two support surfaces **1555***a*, **1555***b* formed in the leading surface **1395**, the leading surface **1395** defines a zig-zag shaped surface. The illustrated third portion **1545**, fourth portion **1550**, and fifth portion **1575** are all linear surfaces with the third portion **1545** obliquely oriented relative to the fourth portion **1550** and the fourth portion **1550** oriented substantially parallel to the fifth portion **1575**. In other embodiments, two or more of the third portion **1545**, the fourth portion **1550**, and the fifth portion **1575** can be substantially parallel relative to each other or obliquely oriented relative to each other. In further embodiments, at least one of the third portion **1545**, the fourth portion **1550**, and the fifth portion **1575** can be a curved surface. In yet further embodiments, the third portion **1545**, the fourth portion **1550**, the fifth portion **1575**, and the two support surface **1555***a*, **1555***b* can be omitted so that the leading surface **1395** is one linear surface. The illustrated first support surface **1555***a* is positioned from the first end **1320** at a first support distance **1560***a* of between about 2 inches and about 2.2 inches. The second support surface **1555***b* is positioned from the first end **1320** at a second support distance **1560***b* of between about 1 inch and about 1.5 inches. In other embodiments, each leading surface **1395** can include one support surface. In other embodiments, each leading surface **1395** can include more than two support surfaces.

(44) In addition, each gullet **1375** defines an oblique angle **1435** between the second portion **1420** and the rotational axis **1350**. The oblique angle **1435** is between about 55 degrees and about 65 degrees. In particular, the oblique angle **1435** is defined between the rotational axis **1350** and a tangent line extending through a midpoint of the second portion **1420**. A first angle **1440** is defined between the first portion **1415** and the second portion **1420** that is between about 120 degrees and

about 135 degrees. In particular, the first angle **1440** is defined between the first portion **1415** and a tangent line extending through a midpoint of the second portion **1420**. In other embodiments with the first portion **1415** defining a curved surface, the first angle **1440** is defined between tangent lines extending through midpoints of the first and second portions **1415**, **1420**. A second angle **1450** is defined between the second portion **1420** and the third portion **1545** that is between about 35 degrees and about 45 degrees. In particular, the second angle **1450** is defined between the third portion **1545** and a tangent line extending through a midpoint of the second portion **1420**. In other embodiments with the third portion **1545** defining a curved surface, the second angle **1450** is defined between tangent lines extending through midpoints of the second and third portions **1420**, **1545**.

(45) As best shown in FIG. **10**, each gullet **1375** defines a gullet height or dimension **1455** extending between the first end **1320** and the bottom surface **1390** in a direction parallel to the rotational axis **1350**. The illustrated gullet height **1455** is between about 3 inches and about 3.2 inches. In other embodiments, the gullet height **1455** can be between about 3.2 inches and about 4 inches, or the gullet height **1455** can be between about 2.5 inches and about 3 inches. As such, a ratio of the sidewall height **1370** over the gullet height **1455** is between about 1.05 and about 1.2. In other embodiments, the ratio can be between about 1.2 and about 2. The illustrated sidewall **1315** also defines a height **1460** extending between the bottom surface **1390** of each gullet **1375** and the second end **1325** of the sidewall **1315**. The height **1460** is between about 0.3 inches and about 0.4 inches. In other embodiments, the height **1460** can be between about 0.4 inches and about 1 inch, or the height **1460** can be between about 0.1 inches and about 0.3 inches. In addition, a maximum gullet width **1465** is defined by a point **1470** on the first portion **1415** and the fifth portion **1575** adjacent the second support surface **1555***b*.

(46) The hole saw **1300** also includes cutting teeth **1475** each having a cutting tip **1480** and seated within one notch **1405**. Each cutting tooth **1475** includes a rake surface **1485** and a relief surface **1490**. A minimum gullet width **1496** extends between each cutting tip **1480** and the corresponding leading surface **1395**. Once the cutting teeth **1475** are coupled to the sidewall **1315**, each gullet **1375** defines an area **1525** between the rake surface **1485**, the trailing surface **1400**, the bottom surface **1390**, and the leading surface **1395**. In the illustrated embodiment, the area **1525** of each gullet **1375** is equal to or greater than 3.2 inches squared. In other embodiments, the area **1525** of each gullet **1375** is between about 2 inches squared and about 4 inches squared. In further embodiments, the area **1525** of each gullet **1375** is between about 3.2 inches squared and about 4 inches squared. As the hole saw **1300** includes three gullets **1375**, the total area of the illustrated gullets **1375** is equal to or greater than 9.6 inches squared. Each gullet **1375** also defines a volume, which is determined by multiplying the area **1525** of each gullet **1375** by the maximum thickness **1365** of the sidewall **1315**. As such, the volume of each gullet **1375** is equal to or greater than 0.256 inches cubed. In other embodiments, the volume of each gullet **1375** can be between about 0.227 inches cubed and about 0.3 inches cubed.

(47) With reference to FIG. **11**, the hole saw **1300** is operable to cut into two workpieces **1530** that defines a workpiece height **1535**. In the illustrated embodiment, the workpiece height **1535** is about 3 inches (e.g., two standard 2 by 4 pieces of lumber). In other embodiments, the workpiece height **1535** can be less than 3 inches. In further embodiments, the workpiece height **1535** can be between about 1.5 inches and about 3 inches. In yet further embodiments, the two workpieces **1530** can be formed as one workpiece with the workpiece height **1535**. In operation, the cutting teeth **1475** cut into the two workpieces **1530**, thereby forming workpiece chips that collect within the gullets **1375**. As the hole saw **1300** bores deeper into the two workpieces **1530**, more workpiece chips are collected within the gullets **1375**. Accordingly, the illustrated gullets **1375** include a particular volume to accommodate the workpiece chips as the hole saw **1300** cuts into the two workpieces **1530**. Once the hole saw **1300** bores through the two workpieces **1530**, a cylindrical workpiece plug will be removed from the cavity **1310** of the hole saw **1300**. In one embodiment, a user can

grip the cylindrical workpiece plug (via their fingers) through the gullets **1375** to axially slide the cylindrical workpiece plug along the rotational axis **1350** and out of the cavity **1310**. In other embodiments, a tool can be inserted into one of the gullets **1375** to be supported on one of the support surfaces **1555***a*, **1555***b* for a user to leverage the cylindrical workpiece plug out of the hole saw **1300**.

(48) FIG. **12** illustrates a planar view of a portion of a hole saw **1600** according to another embodiment. The hole saw **1600** is similar to the hole saw **100** of FIGS. **1-5**; therefore, similar components are designated with similar references numbers plus 1500. At least some differences and/or at least some similarities between the hole saws **100**, **1600** will be discussed in detail below. In addition, components or features described with respect to only one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers. For example, the hole saw **1600** can be sized to bore into the single workpiece **330** of FIG. **5** similar to the hole saw **100**, or the hole saw **1600** can be sized to bore into the two workpieces **1530** of FIG. **11** similar to the hole saw **1300**.

(49) The illustrated hole saw **1600** is moveable about a rotational axis **1650** in a cutting direction **1655** and includes a sidewall **1615** having a first end **1620** and a second end **1625**. The sidewall **1615** also includes gullets **1675** (only one gullet **1675** is illustrated in FIG. **12**) each defining a central longitudinal axis **1680**. Each gullet **1675** includes a notch **1705**, an open end **1685**, a bottom surface **1690** defining a second radius **1745**, a leading surface **1695**, and a trailing surface **1700**. Each trailing surface **1700** includes a trailing surface portion **1710** defining a first radius **1730**, a first portion **1715**, and a second portion **1720** with the trailing surface portion **1710** positioned between the first portion **1715** and the second portion **1720**. The sidewall **1615** also includes three support apertures **1880** (only one is illustrated in FIG. **12**) each positioned at a substantially same location that is circumferentially between adjacent gullets **1675**. In particular, each support aperture **1880** is positioned above the bottom surfaces **1690** of the gullets **1675**. For example, each support aperture **1880** is positioned above the trailing surface portions **1710** of the gullets **1675**. Each support aperture **1880** is also positioned between adjacent gullets **1675** as to not circumferentially overlap with any portion of the gullets **1675**. Each support aperture **1880** is oblong-shaped and includes a longitudinal axis that is substantially perpendicular to the rotational axis **1650**. The support apertures **1880** function similar to the support surfaces illustrated within FIGS. **7-11** for a tool to be inserted into one of the support apertures **1880** to leverage and remove a cylindrical workpiece plug from the hole saw **1600**. In other embodiments, the sidewall **1615** can include fewer than or more than three support apertures **1880** and/or the support apertures **1880** can be located at different positions circumferentially around the sidewall **1615**.

(50) The hole saw **1600** also includes cutting teeth **1775** (only one cutting tooth **1775** is illustrated in FIG. **12**) each having a cutting tip **1780** and seated within one notch **1705**. Each cutting tooth **1775** includes a rake surface **1785** and a relief surface **1790**.

(51) FIG. **13** illustrates a planar view of a portion of a hole saw **1900** according to another embodiment. The hole saw **1900** is similar to the hole saw **100** of FIGS. **1-5**; therefore, similar components are designated with similar references numbers plus 1800. At least some differences and/or at least some similarities between the hole saws **100**, **1900** will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers. For example, the hole saw **1900** can be sized to bore into the single workpiece **330** of FIG. **5** similar to the hole saw **100**, or the hole saw **1900** can be sized to bore into the two workpieces **1530** of FIG. **11** similar to the hole saw **1300**.

(52) The illustrated hole saw **1900** is moveable about a rotational axis **1950** in a cutting direction **1955** and includes a sidewall **1915** having a first end **1920** and a second end **1925**. The sidewall **1915** also includes gullets **1975** (only one gullet **1975** is illustrated in FIG. **13**) each defining a central longitudinal axis **1980**. Each gullet **1975** includes a notch **2005**, an open end **1985**, a bottom

surface **1990** defining a second radius **2045**, a leading surface **1995**, and a trailing surface **2000**. In particular, the second radius **2045** of each bottom surface **1990** extends directly from the trailing surface **2000** to the leading surface **1995**. The sidewall **1915** also includes three support apertures **2180** (only one is illustrated in FIG. **13**) each positioned below the bottom surfaces **1900**. Each support aperture **2180** is also positioned as to circumferentially overlap with one of the gullets **1975**.

(53) The hole saw **1900** also includes cutting teeth **2075** (only one cutting tooth **2075** is illustrated in FIG. **13**) each having a cutting tip **2080** and seated within one notch **2005**. Each cutting tooth **2075** includes a rake surface **2085** and a relief surface **2090**.

(54) FIG. **14** illustrates a planar view of a portion of a hole saw **2200** according to another embodiment. The hole saw **2200** is similar to the hole saw **100** of FIGS. **1-5**; therefore, similar components are designated with similar references numbers plus 2100. At least some differences and/or at least some similarities between the hole saws **100**, **2200** will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers. For example, the hole saw **2200** can be sized to bore into the single workpiece **330** of FIG. **5** similar to the hole saw **100**, or the hole saw **2200** can be sized to bore into the two workpieces **1530** of FIG. **11** similar to the hole saw **1300**.

(55) The illustrated hole saw **2200** is moveable about a rotational axis **2250** in a cutting direction **2255** and includes a sidewall **2215** having a first end **2220** and a second end **2225**. The sidewall **2215** also includes gullets **2275** (only one gullet **2275** is illustrated in FIG. **14**) each defining a central longitudinal axis **2280**. Each gullet **2275** includes a notch **2305**, an open end **2285**, a bottom surface **2290** defining a second radius **2345**, a leading surface **2295**, and a trailing surface **2300**. Each trailing surface **2300** includes a trailing surface portion **2310** defining a first radius **2330**, a first portion **2315**, and a second portion **2320** with the trailing surface portion **2310** positioned between the first portion **2315** and the second portion **2320**. The first portion **2315** of each gullet **2275** defines a third angle **2440** that extends from the notch **2305** to the trailing surface portion **2310** in a direction away from the cutting direction **2255**. As a result, the trailing surface portion **2310** is positioned behind the first portion **2315** relative to the cutting direction **2255**. In other embodiments, the third angle **2440** can be oriented in a direction toward the cutting direction **2255** with at least a portion of the trailing surface portion **2310** positioned behind a portion of the first portion **2315** in a direction relative to the cutting direction **2255**. The sidewall **2215** also includes three support apertures **2480** (only one is illustrated in FIG. **14**) each positioned below a trailing surface portion **2310**.

(56) The hole saw **2200** also includes cutting teeth **2375** (only one cutting tooth **2375** is illustrated in FIG. **14**) each having a cutting tip **2380** and seated within one notch **2305**. Each cutting tooth **2375** includes a rake surface **2385** and a relief surface **2390**.

(57) FIG. **15** illustrates a planar view of a portion of a hole saw **2500** according to another embodiment. The hole saw **2500** is similar to the hole saw **100** of FIGS. **1-5**; therefore, similar components are designated with similar references numbers plus 2400. At least some differences and/or at least some similarities between the hole saws **100**, **2500** will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers. For example, the hole saw **2500** can be sized to bore into the single workpiece **330** of FIG. **5** similar to the hole saw **100**, or the hole saw **2500** can be sized to bore into the two workpieces **1530** of FIG. **11** similar to the hole saw **1300**.

(58) The illustrated hole saw **2500** is moveable about a rotational axis **2550** in a cutting direction **2555** and includes a sidewall **2515** having a first end **2520** and a second end **2525**. The sidewall **2515** also includes gullets **2575** (only one gullet **2575** is illustrated in FIG. **15**) each defining a central longitudinal axis **2580**. Each gullet **2575** includes a notch **2605**, an open end **2585**, a bottom

surface **2590** defining a second radius **2645**, a leading surface **2595**, and a trailing surface **2600**. Each leading surface **2595** includes a support surface **2755** positioned between a third portion **2745** and a fourth portion **2750**.

(59) The hole saw **2500** also includes cutting teeth **2675** (only one cutting tooth **2675** is illustrated in FIG. **15**) each having a cutting tip **2680** and seated within one notch **2605**. Each cutting tooth **2675** includes a rake surface **2685** and a relief surface **2690**.

(60) FIG. **16** illustrates a planar view of a portion of a hole saw **2800** according to another embodiment. The hole saw **2800** is similar to the hole saw **100** of FIGS. **1-5**; therefore, similar components are designated with similar references numbers plus 2700. At least some differences and/or at least some similarities between the hole saws **100**, **2800** will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers. For example, the hole saw **2800** can be sized to bore into the single workpiece **330** of FIG. **5** similar to the hole saw **100**, or the hole saw **2800** can be sized to bore into the two workpieces **1530** of FIG. **11** similar to the hole saw **1300**.

(61) The illustrated hole saw **2800** is moveable about a rotational axis **2850** in a cutting direction **2855** and includes a sidewall **2815** having a first end **2820** and a second end **2825**. The sidewall **2815** also includes gullets **2875** (only one gullet **2875** is illustrated in FIG. **16**) each defining a central longitudinal axis **2880**. Each gullet **2875** includes a notch **2905**, an open end **2885**, a bottom surface **2890** defining a second radius **2945**, a leading surface **2895**, and a trailing surface **2900**. Each trailing surface **2900** includes a trailing surface portion **2910** defining a first radius **2930**, a first portion **2915**, and a second portion **2920** with the trailing surface portion **2910** positioned between the first portion **2915** and the second portion **2920**. The sidewall **2815** also includes three support apertures **3080** (only one is illustrated in FIG. **16**). Each support aperture **3080** defines a greater area than an area of one of the gullets **2875**. Each support aperture **3080** circumferentially overlaps with one of the gullets **2875** so that the leading surface **2895** of the gullet **2875** is positioned forward the support aperture **3080** but a portion of the support aperture **3080** is positioned behind the trailing surface **2900** of the gullet **2875** relative to the cutting direction **2855**.

(62) The hole saw **2800** also includes cutting teeth **2975** (only one cutting tooth **2975** is illustrated in FIG. **16**) each having a cutting tip **2980** and seated within one notch **2905**. Each cutting tooth **2975** includes a rake surface **2985** and a relief surface **2990**.

(63) FIG. **17** illustrates a planar view of a portion of a hole saw **3100** according to another embodiment. The hole saw **3100** is similar to the hole saw **100** of FIGS. **1-5**; therefore, similar components are designated with similar references numbers plus 3000. At least some differences and/or at least some similarities between the hole saws **100**, **3100** will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers. For example, the hole saw **3100** can be sized to bore into the single workpiece **330** of FIG. **5** similar to the hole saw **100** or the hole saw **3100** can be sized to bore into the two workpieces **1530** of FIG. **11** similar to the hole saw **1300**.

(64) The illustrated hole saw **3100** is moveable about a rotational axis **3150** in a cutting direction **3155** and includes a sidewall **3115** having a first end **3120** and a second end **3125**. The sidewall **3115** also includes gullets **3175** (only one gullet **3175** is illustrated in FIG. **17**) each defining a central longitudinal axis **3180**. Each gullet **3175** includes a notch **3205**, an open end **3185**, a bottom surface **3190** defining a second radius **3245**, a leading surface **3195**, and a trailing surface **3200**. Each trailing surface **3200** includes a trailing surface portion **3210** defining a first radius **3230**, a first portion **3215**, and a second portion **3220** with the trailing surface portion **3210** positioned between the first portion **3215** and the second portion **3220**. The sidewall **3115** also includes three support apertures **3380** (only one is illustrated in FIG. **17**). Each support aperture **3380** circumferentially overlaps with one of the gullets **3175** so that a portion of the support aperture

3380 is positioned forward the leading surface **3195** but a portion of the first portion **3215** is positioned behind the support aperture **3380** relative to the cutting direction **3155**.
(65) The hole saw **3100** also includes cutting teeth **3275** (only one cutting tooth **3275** is illustrated in FIG. **17**) each having a cutting tip **3280** and seated within one notch **3205**. Each cutting tooth **3275** includes a rake surface **3285** and a relief surface **3290**.
(66) FIG. **18** illustrates a planar view of a portion of a hole saw **3400** according to another embodiment. The hole saw **3400** is similar to the hole saw **100** of FIGS. **1-5**; therefore, similar components are designated with similar references numbers plus 3300. At least some differences and/or at least some similarities between the hole saws **100**, **3400** will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers. For example, the hole saw **3400** can be sized to bore into the single workpiece **330** of FIG. **5** similar to the hole saw **100**, or the hole saw **3400** can be sized to bore into the two workpieces **1530** of FIG. **11** similar to the hole saw **1300**.
(67) The illustrated hole saw **3400** is moveable about a rotational axis **3450** in a cutting direction **3455** and includes a sidewall **3415** having a first end **3420** and a second end **3425**. The sidewall **3415** also includes three pairs of gullets **3475** (only one pair of gullets **3475** is illustrated in FIG. **18**). Each pair **3475** includes a smaller gullet **3475***a* defining a central longitudinal axis **3480***a* and a larger gullet **3475***b* defining a central longitudinal axis **3480***b*. In the illustrated embodiment, the central longitudinal axes **3480***a*, **3480***b* are substantially parallel; however, in other embodiments, the central longitudinal axes **3480***a*, **3480***b* can be obliquely angled. The smaller gullet **3475***a* includes a notch **3505***a*, an open end **3485***a*, a bottom surface **3490***a* defining a second radius **3545***a*, a leading surface **3495***a*, and a trailing surface **3500***a*. Likewise, the larger gullet **3475***b* includes a notch **3505***b*, an open end **3485***b*, a bottom surface **3490***b* defining a second radius **3545***b*, a leading surface **3495***b*, and a trailing surface **3500***b*. The bottom surface **3490***b* of the larger gullet **3475***b* is positioned closer to the second end **3425** of the sidewall **3415** than the bottom surface **3490***a* of the smaller gullet **3475***a*. In other embodiments, the sidewall **3415** can include fewer or more than three pairs of gullets **3475**. In the illustrated embodiment, the second radius **3545***b* of the larger gullet **3475***b* is greater than the second radius **3545***a* of the smaller gullet **3475***a*. In other embodiments, the second radius **3545***b* of the larger gullet **3475***b* can be equal to or less than the second radius **3545***a* of the smaller gullet **3475***a*.
(68) The hole saw **3400** also includes cutting teeth **3575** (only one pair of cutting teeth **3575** are illustrated in FIG. **18**) each having a cutting tip **3580** and seated within one notch **3505***a*, **3505***b*. Each cutting tooth **3575** includes a rake surface **3585** and a relief surface **3590**.
(69) FIG. **19** illustrates a planar view of a portion of a hole saw **3700** according to another embodiment. The hole saw **3700** is similar to the hole saw **100** of FIGS. **1-5**; therefore, similar components are designated with similar references numbers plus 3600. At least some differences and/or at least some similarities between the hole saws **100**, **3700** will be discussed in detail below. In addition, components or features described with respect to one or some of the embodiments described herein are equally applicable to any other embodiments described herein and can include similar reference numbers. For example, the hole saw **3700** can be sized to bore into the single workpiece **330** of FIG. **5** similar to the hole saw **100**, or the hole saw **3700** can be sized to bore into the two workpieces **1530** of FIG. **11** similar to the hole saw **1300**.
(70) The illustrated hole saw **3700** is moveable about a rotational axis **3750** in a cutting direction **3755** and includes a sidewall **3715** having a first end **3720** and a second end **3725**. The sidewall **3715** also includes gullets **3775** (only one gullet **3775** is illustrated in FIG. **19**) each defining a central longitudinal axis **3780** parallel to the rotational axis **3750**. Each gullet **3775** includes a notch **3805**, an open end **3785**, a bottom surface **3790** defining a second radius **3845**, a leading surface **3795**, and a trailing surface **3800**. Each trailing surface **3800** includes a trailing surface portion **3810** defining a first radius **3830**, a first portion **3815**, and a second portion **3820** with the trailing

surface portion **3810** positioned between the first portion **3815** and the second portion **3820**. The sidewall **3715** also includes three support apertures **3980** (only one is illustrated in FIG. **19**). Each support aperture **3980** defines a longitudinal axis that substantially aligns with one central longitudinal axis **3780** of one gullet **3775**. In particular, each support aperture **3980** is directly below one gullet **3775**.
(71) The hole saw **3700** also includes cutting teeth **3875** (only one cutting tooth **3875** is illustrated in FIG. **19**) each having a cutting tip **3880** and seated within one notch **3805**. Each cutting tooth **3875** includes a rake surface **3885** and a relief surface **3890**.
(72) Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the invention as described. Various features and advantages of the invention are set forth in the following claims.

Claims

1. A hole saw movable about a rotational axis in a cutting direction to cut into a workpiece, the hole saw comprising: a cylindrical body including a sidewall having a first end and a second end opposite the first end, a gullet formed in the sidewall between a leading surface, a trailing surface, and a linear bottom surface of the sidewall, the gullet extending through the first end of the body, and a support aperture formed in the sidewall, the support aperture entirely surrounded by the sidewall, at least one side of the support aperture being parallel to the linear bottom surface of the sidewall, and an entirety of the support aperture being positioned closer than the linear bottom surface of the sidewall to the second end of the sidewall; a cutting tooth coupled to the first end of the sidewall adjacent the gullet; and an end cap coupled to the second end of the sidewall.
2. The hole saw of claim 1, wherein the support aperture has a maximum width measured in a direction perpendicular to the rotational axis, and wherein the maximum width of the support aperture is substantially equal to a maximum width of the gullet.
3. The hole saw of claim 1, wherein the gullet has a height measured in a direction parallel to the rotational axis, and wherein the height of the gullet is greater than half of a height of the sidewall.
4. The hole saw of claim 1, wherein the gullet has a height measured in a direction parallel to the rotational axis, and wherein the height of the gullet is greater than a height of the support aperture.
5. The hole saw of claim 1, wherein the support aperture is at least partially defined by four linear surfaces of the sidewall.
6. The hole saw of claim 1, wherein a shortest distance between the gullet and the support aperture measured parallel to the rotational axis is less than a shortest distance between the support aperture and the second end of the sidewall measured parallel to the rotational axis.
7. A hole saw movable about a rotational axis in a cutting direction to cut into a workpiece, the hole saw comprising: a cylindrical body including a sidewall having a first end and a second end opposite the first end, a gullet formed in the sidewall between a first surface, a second surface, and a third surface of the sidewall, the third surface extending between the first and second surfaces, the gullet extending through the first end of the sidewall, and a polygonal opening formed in the sidewall, the polygonal opening entirely surrounded by the sidewall and positioned between the third surface and the second end of the sidewall; a cutting tooth coupled to the first end of the sidewall adjacent the second surface of the gullet; and an end cap coupled to the second end of the sidewall, wherein the gullet has a height measured in a direction parallel to the rotational axis, and wherein the height of the gullet is greater than a height of the polygonal opening.
8. The hole saw of claim 7, wherein the polygonal opening has a rounded parallelogram shape.
9. The hole saw of claim 7, wherein the polygonal opening includes a first side, a second side, a third side, and a fourth side, and wherein the first side of the polygonal opening is parallel with the third surface of the sidewall.

10. The hole saw of claim 9, wherein the first and second sides of the polygonal opening are parallel, and wherein the third and fourth sides of the polygonal opening are parallel.
11. The hole saw of claim 9, wherein the first, second, third, and fourth sides of the polygonal opening are connected by rounded corners.
12. The hole saw of claim 9, wherein the polygonal opening has a height measured between the first and second sides and parallel to the rotational axis, and wherein the height of the polygonal opening is greater than a shortest distance between the third surface of the sidewall and the first side of the polygonal opening measured parallel to the rotational axis.
13. The hole saw of claim 9, wherein the third surface of the sidewall is linear, and wherein the first side of the polygonal opening is linear.
14. The hole saw of claim 13, wherein a first arcuate surface is positioned between the first and third surfaces of the sidewall, and wherein a second arcuate surface is positioned between the second and third surfaces of the sidewall.
15. A hole saw movable about a rotational axis in a cutting direction to cut into a workpiece, the hole saw comprising: a cylindrical body including a sidewall having a first end and a second end opposite the first end, a gullet formed in the sidewall between a leading surface, a trailing surface, and a bottom surface of the sidewall, the gullet extending through the first end of the sidewall, the gullet having a maximum width measured between the leading surface and the trailing surface in a direction perpendicular to the rotational axis, and a support aperture formed in the sidewall, the support aperture having a first side, a second side opposite the first side, a linear third side extending between the first side and the second side, and a maximum width measured between the first side and the second side in the direction perpendicular to the rotational axis, the maximum width of the support aperture being substantially equal to the maximum width of the gullet; a cutting tooth coupled to the first end of the sidewall adjacent the gullet; and an end cap coupled to the second end of the sidewall.
16. The hole saw of claim 15, wherein the support aperture also has a fourth side extending between the first side and the second side.
17. The hole saw of claim 16, wherein the third side of the support aperture is a side of the support aperture nearest the gullet, the fourth side of the support aperture is a side of the support aperture nearest the second end of the sidewall, and the support aperture is devoid of material between the third and fourth sides.
18. The hole saw of claim 15, wherein the third side of the support aperture is positioned closer than the bottom surface of the sidewall to the second end of the sidewall.
19. The hole saw of claim 15, wherein the gullet has a height measured parallel to the rotational axis, the support aperture has a height measured parallel to the rotational axis, and the height of the gullet is greater than the height of the support aperture.
20. The hole saw of claim 15, wherein a width of the gullet narrows from the maximum width as the gullet approaches the first end of the sidewall.